Don Bosco Institute of Technology, Mumbai

(An Autonomous Institute Affiliated to the University of Mumbai)

Mumbai - 400070

Department of Electronics & Telecommunication Engineering

IoT-Enabled Autonomous Guided Vehicle (AGV) for Smart Warehouse Automation

A Report Submitted

in partial fulfillment of the requirements for

COMMUNITY ENGAGEMENT PROJECT 3

Course Code: 25ET5CEP03

Semester V, Academic Year 2026-27

Submitted by

Name of Student	Roll No.	Student ID
Yash Khanvilkar	26	2024110052
Rohit Kumar	28	2024110020
Harshit Patil	37	2024110044
Tirth Shetty	46	2024110055

Supervised by

Prof. Pratibha Dumane

AY 2026-27

CERTIFICATE

This is to certify that the Community Engagement Project 3 entitled "IoT-Enabled Autonomous Guided Vehicle (AGV) for Smart Warehouse Automation" is a bona fide work carried out by the following students of Third Year Electronics & Telecommunication Engineering, Semester V, during Academic Year 2026-27, in partial fulfillment of the requirements of the course Community Engagement Project 3 (25ET5CEP03).

Student Name	**Roll No.**	**Signature**
Yash Khanvilkar	26	
Rohit Kumar	28	
Harshit Patil	37	
Tirth Shetty	46	

Faculty Mentor	**Head of Department**	**Principal**
Prof. Pratibha Dumane	Dr. Madhavi Pednekar	Dr. Sudhakar Mande

COMMUNITY ENGAGEMENT PROJECT 3 REPORT APPROVAL

This report titled "IoT-Enabled Autonomous Guided Vehicle (AGV) for Smart Warehouse Automation" submitted by:

1. Yash Khanvilkar
2. Rohit Kumar
3. Harshit Patil
4. Tirth Shetty

is approved for completion of Community Engagement Project 3 (25ET5CEP03), Semester V, Academic Year 2026-27, Department of Electronics & Telecommunication Engineering.

Examiners

1. _____

2. _____

Date: _____

Place: Kurla, Mumbai

DECLARATION

We hereby declare that the project report titled "IoT-Enabled Autonomous Guided Vehicle (AGV) for Smart Warehouse Automation" is our original work carried out under the guidance of Prof. Pratibha Dumane. The work has not been submitted elsewhere for the award of any degree or academic credit. All sources of information, data, images, code, standards, software libraries and published material used in this report have been appropriately acknowledged. Any external or open-source code, datasets, images or AI-assisted tools used in the final implementation will be acknowledged in the relevant sections and references.

Student Name	**Roll No.**	**Signature**
Yash Khanvilkar	26	
Rohit Kumar	28	
Harshit Patil	37	
Tirth Shetty	46	

VISION, MISSION AND PROGRAMME SPECIFIC OUTCOMES

Institute Vision

The Don Bosco Institute of Technology (DBIT) will be known to have an innovative, enjoyable and holistic learning environment that transforms individuals into socially conscious citizens the Don Bosco Way, and will lead in research and entrepreneurship in the area of sustainable technologies.

Institute Mission

- To create future engineers who work with honesty and integrity and excel in the use of technology for the benefit of the underprivileged.
- To train engineers to be innovative problem solvers and entrepreneurs who engage in research and life-long learning.
- To provide a diverse and stimulating environment for staff and students to grow holistically.

Department Vision

Department of Electronics and Telecommunication Engineering envisions technical professionals with principles of Don Bosco contributing to research, technology, society and sustainable development.

Department Mission

- To produce technically competent, employable and socially responsible graduates with professional ethics.
- To facilitate creativity, research, innovation, teamwork and entrepreneur skills to meet academic, industrial and societal needs.

Programme Specific Outcomes (PSOs)

1. **PSO 1:** Ability to utilise the concepts of Electronics & Telecommunication Engineering and allied fields to attain industry-ready competence.
2. **PSO 2:** Ability to acquire and apply pertinent engineering skills to analyse problems and provide design solutions in the field of Communication, Signal Processing, Embedded Systems.

ACKNOWLEDGEMENTS

We express our sincere gratitude to the Management of The Bombay Salesian Society's Don Bosco Institute of Technology, Mumbai, for providing the academic environment and facilities required to undertake this Community Engagement Project 3. We are thankful to the Principal, Dr. Sudhakar Mande, and the Head of the Department of Electronics & Telecommunication Engineering, Dr. Madhavi Pednekar, for their support and encouragement.

We are especially grateful to our faculty mentor, Prof. Pratibha Dumane, for her guidance, technical suggestions and continuous support during the planning, design, implementation and documentation of the IoT-Enabled Autonomous Guided Vehicle for Smart Warehouse Automation. We also acknowledge the support of the EXTC Department and the laboratory staff.

The team reviewed publicly available information about potential logistics stakeholders, including ColdStar Logistics Pvt. Ltd., TCF Logistics Pvt. Ltd. and LP Logiscience LLP. The supplied logbook does not record direct meetings, interviews or feedback from these organisations; therefore, no stakeholder endorsement or direct feedback is claimed in this report. We thank our classmates and all individuals who contributed to discussions and learning during the project.

ABSTRACT

Warehouse operations involve repetitive movement of materials between storage, sorting and dispatch areas. Manual transportation can require substantial operator time and may provide limited real-time visibility of vehicle movement and delivery status. This project presents the design and preliminary development of a low-cost, IoT-enabled Autonomous Guided Vehicle (AGV) intended as a proof of concept for predefined-route material transportation in structured warehouse environments.

The prototype uses an ESP32 development board as its main controller. Three infrared sensors are used for black-line navigation, an MFRC522/RC522 RFID reader is provided for identifying predefined warehouse checkpoints, and an HC-SR04 ultrasonic sensor is used for local obstacle detection. A servo-mounted ultrasonic arrangement supports a scanning concept. Two geared DC motors are driven in a differential-drive configuration through a TB6612FNG motor driver. The power subsystem consists of four 18650 cells arranged as a 2S2P pack, a 2S battery-management system and an LM2596 buck converter. Wi-Fi connectivity supports a Blynk monitoring interface and a Google Apps Script / Google Sheets event-logging layer. An ESP32-CAM is included as a visual-monitoring extension.

Preliminary functional validation demonstrates line following on the developed guidepath. A Blynk dashboard interface test is documented; however, the supplied technical report explicitly distinguishes dashboard connectivity from end-to-end delivery validation. RFID destination logic, obstacle-response threshold and field testing, Google Sheets event logging and ESP32-CAM streaming remain subject to integrated validation. No quantitative accuracy, speed, RFID range, battery endurance or latency results were supplied. The prototype therefore demonstrates the feasibility of combining guided navigation and monitoring functions, while further testing and direct stakeholder consultation are required before any operational or industrial deployment claim can be made.

CONTENTS

CERTIFICATE ii

COMMUNITY ENGAGEMENT PROJECT 3 REPORT APPROVAL iii

DECLARATION iv

VISION, MISSION AND PROGRAMME SPECIFIC OUTCOMES v

ACKNOWLEDGEMENTS vi

ABSTRACT vii

LIST OF FIGURES xi

LIST OF TABLES xii

1 INTRODUCTION AND COMMUNITY NEED ANALYSIS 1

1.1 Background and Project Motivation 1
1.2 Community Need / Societal Problem Identified 1
1.3 Stakeholder Identification and Requirement Gathering 1
1.4 Problem Statement 2
1.5 Project Objectives 3
1.6 Scope and Limitations 3
1.7 Expected Community / Societal Benefit 3

2 LITERATURE REVIEW AND PROBLEM ANALYSIS 4

2.1 Literature Survey 4
2.2 Existing Solutions / Technologies 6
2.3 Comparative Analysis and Identified Gap 6
2.4 Refined Problem Requirements / Performance Criteria 7

3 PROJECT PLANNING, SDG AND KAP ALIGNMENT 9

3.1 Project Plan and Milestones 9
3.2 Resources and Tools 9
3.3 Relevant Sustainable Development Goal(s) 10
3.4 Knowledge and Attitude Profiles (KAPs) 11
3.5 Risks, Safety and Ethical Considerations 11

4 SYSTEM DESIGN AND METHODOLOGY 12

4.1 Proposed Solution / System Overview 12
4.2 System Architecture / Block Diagram / Workflow 13

4.3 Design Methodology 13
4.4 Hardware / Software / Communication / Tools Selection 14
4.5 Design Constraints and Trade-offs 15

5 IMPLEMENTATION AND DEVELOPMENT 17
5.1 Implementation Details 17
5.2 Module-wise / Phase-wise Development 17
5.2.1 Mechanical Platform and Drive 17
5.2.2 Line-following Subsystem 17
5.2.3 RFID Checkpoint Subsystem 18
5.2.4 Obstacle Detection and Servo Subsystem 18
5.2.5 IoT Monitoring and Camera Subsystems 18
5.2.6 Exhibition Setup and Prototype Demonstration 19
5.3 Integration and Troubleshooting 20
5.4 Cost / Resource Summary 21

6 TESTING, RESULTS AND VALIDATION 23
6.1 Test Plan / Validation Method 23
6.2 Results and Performance Analysis 24
6.3 Stakeholder Feedback and Validation 25
6.4 Achievement of Project Objectives 25

7 SUSTAINABILITY, SOCIETAL IMPACT AND ENGINEERING CONSIDERATIONS 27
7.1 Sustainability Assessment 27
7.2 Societal Relevance, Inclusiveness and Accessibility 27
7.3 Ethical, Safety and Professional Responsibility 27
7.4 Complex Engineering / Problem-Solving Evidence 28
7.5 Community Impact / Potential Impact 28

8 CONCLUSION, REFLECTION AND FUTURE WORK 29
8.1 Conclusion 29
8.2 Limitations 29
8.3 Future Improvements and Scalability 29
8.4 Team Reflection and Learning Outcomes 30
8.5 Individual Contribution of Team Members 30

REFERENCES 32

APPENDICES 34
Appendix A – Stakeholder Interaction / Community Evidence 34
Appendix B – Testing / Validation Evidence 34
Appendix C – Technical Documentation 35

Appendix D – Project Demonstration / Repository Links 36
Appendix E – Course Outcomes, CO-PO-PSO Mapping and Evidence of CO Achievement 36

LIST OF FIGURES

4.1 Project architecture and functional block diagram of ATLAS, showing the user-interface, control, sensing, actuation and power subsystems. 13

5.1 Developed AGV prototype used for preliminary functional validation. 17
5.2 Blynk IoT dashboard interface documented for AGV status monitoring. The screenshot demonstrates the interface test only. 18
5.3 Front view of the assembled AGV. 19
5.4 Top view of the AGV hardware arrangement. 19
5.5 ATLAS carrying the demonstration payload during the exhibition setup. 20
5.6 Side view of the AGV placed on the black guide track at the exhibition. 20
5.7 Front view showing the ultrasonic sensing and upper camera arrangement. 20
5.8 AGV and demonstration track with the project presentation material. 20

LIST OF TABLES

1.1 Potential stakeholders and requirement-gathering status 2

2.1 Selected literature relevant to the proposed AGV 4
2.2 Refined requirements and validation criteria 7

3.1 Project milestones and progress status 9
3.2 Sustainable Development Goal alignment 10
3.3 Knowledge and Attitude Profiles demonstrated through the project 11

4.1 Main hardware and software components 14

5.1 Documented integration challenges and follow-up actions 20
5.2 Estimated bill of materials based on the project poster 21

6.1 Test plan and current validation status 23
6.2 Stakeholder feedback and validation status 25
6.3 Objective-wise achievement status 25

8.1 Individual contribution of team members 30
8.2 Publicly available stakeholder information recorded in the logbook 34
8.3 Course outcomes and project evidence 36
8.4 CO-PO-PSO mapping justification 38

1. INTRODUCTION AND COMMUNITY NEED ANALYSIS

1.1 Background and Project Motivation

Warehouses and logistics facilities require the movement of goods between receiving, storage, picking, sorting and dispatch locations. These internal movements are often repetitive and may be carried out manually using trolleys or other material-handling equipment. Manual movement can consume operator time, create opportunities for handling errors and make it difficult to obtain a consistent digital record of vehicle status and delivery events. Research on warehouse automation describes the increasing use of automated guided vehicles (AGVs) and autonomous mobile robots to improve internal logistics and reduce repetitive handling activities [1], [2], [3], [4].

An AGV that follows a predefined route is a practical entry point for structured environments because it uses a physical guidepath rather than requiring a complex global-localisation system. The present project extends a conventional line-following robot by combining guidepath navigation with RFID checkpoint identification, ultrasonic obstacle sensing, remote status monitoring and a camera-based monitoring extension. The intention is to develop an accessible educational prototype that demonstrates how embedded systems and IoT technologies can support warehouse material movement.

1.2 Community Need / Societal Problem Identified

The community need considered in this project is the repetitive movement of goods within small and medium-sized warehouses and logistics facilities. Warehouse operators and material-handling personnel may benefit from a compact guided vehicle that can repeatedly travel along a marked route, identify designated zones and provide basic visibility of its operating state. A low-cost demonstrator can help students and prospective users understand the feasibility and limitations of warehouse automation.

The project logbook identifies ColdStar Logistics Pvt. Ltd. and TCF Logistics Pvt. Ltd. in Kurla West, and LP Logiscience LLP with a Kurla office and listed warehouse facilities in Bhiwandi, as potential stakeholders through desk research. The logbook explicitly states that no direct interaction has been confirmed. Accordingly, the problem context in this report is based on desk research and the project team's defined use case, not on completed interviews, field observations or measured stakeholder requirements.

1.3 Stakeholder Identification and Requirement Gathering

Potential stakeholders were identified by reviewing publicly available company information on 15 October 2026. This exercise was used to identify organisations operating in logistics or warehousing that may be suitable for future consultation. No company employee was contacted and no site visit or formal requirement-gathering exercise is documented.

Table 1.1: Potential stakeholders and requirement-gathering status

Stakeholder / Organization	Mode of Interaction	Key Need / Requirement	Evidence Reference
ColdStar Logistics Pvt. Ltd., Kurla West	Online review of public website; no direct interaction	Public information indicates temperature-controlled warehousing, inventory management and shipment visibility as relevant logistics functions. These are potential use-case areas, not interview feedback.	Project logbook, stakeholder record, 15 Oct. 2026; website listed in Appendix A.
TCF Logistics Pvt. Ltd., Kurla West	Online review of public website; no direct interaction	Public information lists warehouse space and locations including Bhiwandi and JNPT. Potential relevance is internal material movement and warehouse automation.	Project logbook, stakeholder record, 15 Oct. 2026; website listed in Appendix A.
LP Logiscience LLP, Kurla office / Bhiwandi facilities	Online review of public contact page and facility locations; no direct interaction	Publicly listed warehouse facilities identify a potential setting for future AGV consultation and demonstration.	Project logbook, stakeholder record, 15 Oct. 2026; website listed in Appendix A.

1.4 Problem Statement

To develop a low-cost, compact and traceable material-transport prototype for a structured warehouse environment that follows a predefined guidepath, recognises designated RFID checkpoints, detects nearby obstacles and provides remote status visibility. The proposed solution is an ESP32-based Autonomous Guided Vehicle (AGV) proof of concept that combines line-following navigation, checkpoint identification, local obstacle sensing and IoT-enabled monitoring. The report distinguishes the functions demonstrated on the prototype from those that still require complete integrated validation.

1.5 Project Objectives

The specific objectives of the project are:

1. To design and assemble a compact AGV prototype for predefined-path warehouse material transportation.
2. To implement black-line navigation using three infrared sensors and a differential-drive motor arrangement.
3. To integrate an MFRC522/RC522 RFID reader for identification of predefined warehouse checkpoints.
4. To integrate an HC-SR04 ultrasonic sensor and a servo-mounted scanning concept for obstacle awareness and motor-stop response.
5. To provide Wi-Fi-based status monitoring through Blynk and timestamped event logging through Google Apps Script and Google Sheets.
6. To incorporate an ESP32-CAM subsystem for visual monitoring and delivery-image capture.

1.6 Scope and Limitations

The project scope includes the mechanical assembly of a compact mobile chassis, ESP32-based local control, three-sensor line following, RFID hardware integration, ultrasonic obstacle sensing, differential-drive actuation, Wi-Fi-based monitoring architecture, event-logging architecture and an ESP32-CAM visual-monitoring extension. The prototype is intended for a controlled demonstration track with predefined routes and checkpoint locations.

The current implementation is not a free-navigating autonomous mobile robot. It depends on a physical guidepath and does not include dynamic route planning, fleet coordination, industrial-grade safety certification, validated payload capacity or a production warehouse-management-system interface. The supplied technical report confirms preliminary line-following validation, while RFID destination logic, obstacle-response threshold and field testing, end-to-end IoT logging and camera streaming require further integrated tests. No quantitative performance values or direct stakeholder validation were supplied.

1.7 Expected Community / Societal Benefit

The expected benefit is to demonstrate a low-cost approach to guided internal material movement and improve the visibility of a vehicle's operating state through basic IoT monitoring. In a future controlled deployment, the system could be assessed using measurable indicators such as route-completion rate, number of successful checkpoint reads, obstacle-stop response, operator intervention count, event-log completeness, battery operating time and time required to move a test load. These benefits are potential outcomes only; time savings, labour reduction and safety improvement have not yet been measured in a real warehouse environment.

2. LITERATURE REVIEW AND PROBLEM ANALYSIS

2.1 Literature Survey

The project team reviewed 20 research papers and selected at least ten relevant studies for comparison with the proposed prototype. The selected literature covers warehouse AGV deployment, RFID-based line-following robots, ESP32-based warehouse automation, small-scale industrial AGVs, navigation strategies, dynamic path planning, dispatching, collision prevention and robot-assisted material handling. The review is used to identify suitable low-cost design choices and the limitations that remain when a laboratory prototype is compared with a deployed warehouse system.

Table 2.1: Selected literature relevant to the proposed AGV

Ref.	Existing Work / Technology	Approach / Method	Key Findings	Limitation / Gap
[1]	Kubasakova et al. (2024), AGV implementation in warehouse processes	Warehouse case study using AGV/AMR, RFID and sensors with route and cycle analysis.	The supplied review reports replacement of repetitive human work by two AGVs and an eight-month payback in the studied case.	Case-study economics depend on the deployment context and cannot be transferred directly to this low-cost prototype.
[14]	Pandit et al. (2026), RFID-based autonomous line-following robot for warehouse inventory management	IR line following, RC522 RFID, ESP32 and Wi-Fi; stop-resume RFID scan approach.	Reports successful line following and improved RFID reading when the robot pauses for scanning.	Focuses on the RFID line-following combination; broader monitoring and integrated visual functions remain a separate system-level consideration.
[15]	Sahu et al. (2026), autonomous warehouse robot with priority-based sorting	ESP32, IR array, ultrasonic sensing, PID / finite-state-machine control and Wi-Fi.	The supplied literature review reports 91.5% line accuracy and 0.32 s obstacle response.	Results are reported by the reviewed study and are not measurements of the present prototype.

Ref.	Existing Work / Technology	Approach / Method	Key Findings	Limitation / Gap
[16]	Katwate et al. (2024), AGV for small-scale industry	Line following, ultrasonic obstacle detection and autonomous movement.	Reports smooth navigation and accurate positioning in a small-scale industrial prototype.	The supplied review does not provide a numerical accuracy value; direct comparison requires a common test protocol.
[2]	Ellithy et al. (2024), AGV and Industry 4.0 in warehouses	Literature analysis and framework for flexible warehouse automation.	Explains the role of AGVs within connected and flexible warehouse systems.	Broad framework; implementation cost and performance depend on the facility.
[3]	Staffenova et al. (2023), AGVs in internal logistics	Review of AGV use in production-company internal logistics.	Describes how AGVs support repetitive movement and material flow.	The benefits depend on route design and production layout.
[4]	Alam et al. (2023), review and challenges of AGVs	Review of AGV technologies and implementation challenges.	Identifies navigation, control, safety and integration as important design concerns.	Review-level discussion; does not validate the present low-cost design.
[5]	Ho et al. (2025), reinforcement-learning AGV path planning	Reinforcement learning for dynamic path planning in smart logistics.	Shows the research direction toward adaptive routing in changing environments.	More computationally demanding than a fixed-guidepath educational prototype.
[6]	Wang et al. (2024), dynamic multi-AGV path planning	Time-reuse strategy for dynamic path planning of multiple AGVs.	Addresses coordination and route efficiency in multi-vehicle settings.	Multi-AGV scheduling is outside the scope of the present single-vehicle prototype.
[7]	Nage (2022), AGV design and methodology	Design-oriented study of an automated guided vehicle.	Provides background on AGV architecture and implementation methodology.	The available review does not provide directly comparable performance data.

Ref.	Existing Work / Technology	Approach / Method	Key Findings	Limitation / Gap
[8]	Bourafa and Siegfried (2023), AGV dispatching and navigation	Review of dispatching systems and navigation concepts.	Highlights the importance of navigation and task allocation in AGV systems.	Dispatching/fleet-level functions are not implemented in this project.
[9]	Aizat et al. (2023), AGV navigation approaches	Survey of navigation methods for dynamic surroundings.	Compares navigation approaches and the challenges posed by changing environments.	Advanced dynamic navigation is beyond the fixed-line route used here.
[11]	Hadisujoto et al. (2023), autonomous warehouse robot	3D-printed warehouse robot using mecanum wheels and a robotic arm.	Demonstrates an alternative approach to autonomous goods handling and manipulation.	More complex mechanical hardware than the fixed-route two-motor AGV developed here.

2.2 Existing Solutions / Technologies

Existing warehouse material-handling approaches include manual trolleys and forklifts, fixed conveyor systems, line-guided AGVs and more flexible autonomous mobile robots. Manual systems are simple to introduce but depend on operator availability and may provide limited automatic traceability. Conveyors are effective for fixed, high-volume transfer routes but require a fixed installation and may not provide flexible point-to-point movement. Conventional line-guided AGVs use a physical route marker such as a line, magnetic strip or other guidepath and can provide repeatable movement in structured environments [3], [7], [16].

More advanced AGVs and AMRs can combine RFID, machine vision, LiDAR, wireless communication and dynamic route planning. Research also explores reinforcement-learning methods and coordinated multi-vehicle scheduling for changing warehouse conditions [5], [6], [9]. Such systems can provide more autonomy but typically require greater sensing, computation, integration effort and cost. The present project selects a predefined guidepath and affordable embedded components to keep the prototype accessible and reproducible.

2.3 Comparative Analysis and Identified Gap

The comparison below is qualitative and reflects the broad characteristics described in the supplied literature; it is not a measured benchmark of commercial products.

- **Manual handling:** low initial technology requirement but dependent on people for every movement and limited automatic event logging.
- **Conveyor systems:** suitable for continuous fixed routes, but less adaptable when destinations

or floor layouts change.

- **Line-guided AGV:** provides repeatable movement on a marked path at comparatively low navigation complexity, but is constrained by the physical guidepath.
- **Advanced AMR:** can support more flexible navigation and route planning, but may require higher-cost perception and localisation hardware and more complex software.

The identified project gap is the need for an educational, low-cost platform that combines line-based navigation, RFID destination identification, local obstacle sensing, a remote monitoring interface, event logging and visual monitoring while retaining a relatively simple control architecture. The work therefore focuses on system-level integration rather than claiming a new line-following or sensing principle.

2.4 Refined Problem Requirements / Performance Criteria

The requirements below translate the proposed use case into functional criteria. Numerical thresholds for accuracy, speed, distance, latency and endurance were not provided in the supplied documents; these values must be established during controlled tests rather than invented in the report.

Table 2.2: Refined requirements and validation criteria

Requirement / Criterion	Type	Target / Expected Value	Source / Stakeholder
Black-line guidepath tracking	Technical	AGV follows the marked route using three IR sensors; functional demonstration on the prepared track.	Project objectives; laboratory guidepath.
Checkpoint identification	Technical / User	Read a valid RFID tag and associate it with a predefined warehouse zone.	Project objectives; future warehouse operator consultation.
Obstacle awareness	Technical / Safety	Measure forward distance and initiate a motor-stop response when the configured threshold is crossed.	Project objectives; safe demonstration-track operation.
Remote status visibility	User / Technical	Display selected operating status through a Blynk dashboard when Wi-Fi is available.	Project objectives; future warehouse operator consultation.

Event record	Technical / User	Record selected events with timestamp, tag ID and status in Google Sheets.	Project objectives; future inventory / operations workflow.
Visual monitoring	User / Technical	Provide camera-based observation using ESP32-CAM; validate stream and image capture.	Project objectives; future operator supervision.

3. PROJECT PLANNING, SDG AND KAP ALIGNMENT

3.1 Project Plan and Milestones

The logbook gives approximate weekly progress records. The activities and dates below preserve that status: the logbook explicitly notes that several dates are approximate recollections and that week-specific test results were not supplied.

Table 3.1: Project milestones and progress status

Milestone / Activity	Planned Period	Actual Status / Completion	Remarks
Literature review and project scope	27 Jul–2 Aug 2026	Review of 20 papers recorded; four relevant studies selected.	Scope and architecture definition continued.
IR sensor testing and line following	3 Aug–20 Sep 2026	Line-following work carried out; functional line following demonstrated on the developed guidepath.	Calibration and black/white detection required attention; no numerical accuracy data supplied.
RFID module integration	17–23 Aug 2026	RFID hardware integration reported complete; destination logic remained under system testing.	Read reliability and destination logic require recorded validation.
Motor and drive platform	24–30 Aug 2026	Differential-drive platform with two geared DC motors assembled.	No week-specific speed or torque measurements supplied.
IoT monitoring and event logging	21–30 Sep 2026	Monitoring / logging architecture defined; Blynk interface test documented.	Google Sheets end-to-end logging validation not supplied.
Obstacle sensing	8–15 Oct 2026	Ultrasonic sensor integration reported complete.	Threshold setting and field-response testing remained in progress.
Camera integration and full validation	20–27 Oct 2026	ESP32-CAM included as a visual-monitoring extension.	Streaming validation and end-to-end delivery test results not supplied.

3.2 Resources and Tools

The main resources used or specified in the project documentation are summarised below. The listed prices are estimates copied from the project poster; actual purchase costs and the final battery configuration must be verified against receipts.

- **Hardware:** ESP32 development board, three IR line sensors, MFRC522/RC522 RFID reader and tags, HC-SR04 ultrasonic sensor, SG90 servo, TB6612FNG motor driver, two TT geared DC motors, buzzer, ESP32-CAM, four 18650 cells, 2S BMS, LM2596 buck converter and mobile chassis.
- **Software / platforms:** Arduino IDE / embedded C-C++, Blynk IoT, Google Apps Script and Google Sheets.
- **Communication:** Wi-Fi for dashboard updates and event-log communication; SPI for RC522 interface as applicable to the selected module.
- **Development resources:** laboratory tools, wiring, mounting hardware, track material, testing instruments and technical datasheets.

3.3 Relevant Sustainable Development Goal(s)

The project is primarily aligned with Sustainable Development Goal (SDG) 9: Industry, Innovation and Infrastructure, which promotes resilient infrastructure, inclusive and sustainable industrialisation, and innovation. The project is also relevant to SDG 12: Responsible Consumption and Production, which focuses on efficient use of resources, waste reduction and more sustainable production and consumption practices. For this prototype, SDG 12 is discussed as a potential contribution through organised material movement and operational visibility; no reduction in material use, energy consumption or waste has been measured.

Table 3.2: Sustainable Development Goal alignment

SDG	Relevance to Project	Evidence / Project Activity
SDG 9: Industry, Innovation and Infrastructure	Demonstrates a low-cost embedded automation prototype for structured warehouse transport and supports learning about industrial digitalisation.	ESP32-based AGV architecture, line following, RFID checkpoint concept and IoT monitoring interface.
SDG 12: Responsible Consumption and Production	Potentially supports more organised internal material movement and operational visibility, which may help reduce avoidable handling and process inefficiency when validated in practice.	Defined material-transport use case and planned event logging; no measured material or energy savings are claimed.

3.4 Knowledge and Attitude Profiles (KAPs)

Table 3.3: Knowledge and Attitude Profiles demonstrated through the project

KAP / Attribute	How Demonstrated	Evidence Reference
Engineering knowledge	Application of embedded control, IR sensing, RFID, ultrasonic ranging, motor driving, power regulation and Wi-Fi communication concepts.	Chapters 4–6; technical paper.
Teamwork and collaboration	Shared planning, literature review, hardware integration, progress presentations and documentation.	Project logbook; Progress Seminar 1 and 2.
Problem solving and lifelong learning	Iterative sensor calibration, module-level testing, literature review and staged integration.	Logbook weekly progress record; Chapter 5.
Professional communication	Technical paper, seminar presentations and structured report preparation.	Supplied paper and seminar slides.
Ethics and responsibility	Explicitly separating desk research from direct stakeholder feedback and avoiding unsupported performance claims.	Logbook stakeholder record; Chapters 6 and 7.
Sustainability awareness	Consideration of the potential role of automation in efficient warehouse material handling.	SDG alignment in Section 3.3.

3.5 Risks, Safety and Ethical Considerations

The prototype uses rechargeable lithium-ion cells, moving motors and wheels, exposed wiring and electronic sensors. The battery pack must be assembled and charged using a suitable 2S protection/charging arrangement; wiring polarity and current ratings must be checked before energising the system. The vehicle should be tested on a clear, controlled track, with hands kept away from rotating wheels and the emergency-stop function checked before each run. Ultrasonic obstacle sensing is a basic local detection mechanism and must not be treated as an industrial personnel-safety system.

The ESP32-CAM may capture identifiable people. Any real-site camera test should be performed only with appropriate permission, informed consent where required, restricted access and limited of captured images. RFID tags used for the demonstration should identify test checkpoints rather than personal identities. Remote status transmission depends on network availability; the primary navigation and local motor-stop decision should remain local to the ESP32. The device is proof of concept and is not certified for deployment around workers or forklifts.

4. SYSTEM DESIGN AND METHODOLOGY

4.1 Proposed Solution / System Overview

The proposed system is a compact two-motor AGV designed to navigate along a predefined black guidepath in a structured warehouse-like environment. An ESP32 development board acts as the main controller and receives inputs from three IR line sensors, an RFID reader and an ultrasonic distance sensor. Based on the sensor state, the controller issues differential motor commands through a TB6612FNG motor driver. RFID tags placed at selected checkpoints are intended to identify warehouse zones. Obstacle distance is monitored using an HC-SR04 sensor, with a servo used for the scanning concept.

The ESP32's Wi-Fi interface provides a connection to a Blynk dashboard and a Google Apps Script / Google Sheets event-logging layer. An ESP32-CAM provides an additional visual-monitoring function. Navigation and safety decisions are kept local to the vehicle so that basic movement and obstacle-stop logic do not depend on cloud availability.

The overall operation is divided into five cooperating stages. First, the IR sensing stage continuously detects the contrast between the black guide line and the light track surface. Second, the ESP32 interprets the sensor pattern and selects the corresponding steering action. Third, the motor-driver stage controls the left and right geared motors independently, allowing the vehicle to move straight or correct its direction. Fourth, the RFID subsystem is intended to identify a checkpoint when the vehicle passes over a tagged warehouse location. Finally, the monitoring stage sends selected operating information to the dashboard and event-logging service when Wi-Fi connectivity is available.

Obstacle awareness is treated as a local supervisory function. The HC-SR04 measures the distance to an object in front of the vehicle, while the servo-mounted arrangement provides a scanning concept for checking different directions. When the configured obstacle condition is detected, the controller can issue a motor-stop and warning command. The report does not claim a verified autonomous detour or obstacle-avoidance manoeuvre because the supplied test evidence does not establish repeatable end-to-end avoidance.

The design uses a layered chassis to separate the battery and drive hardware from the controller, sensing and monitoring electronics. A 2S2P battery arrangement is described in the technical documentation, with a 2S BMS for cell protection and an LM2596 buck converter for regulated low-voltage electronics. This arrangement supports a compact mobile demonstration platform, but actual operating time and load capacity have not been measured. The architecture is modular so that the navigation, RFID, obstacle sensing and IoT functions can be tested individually before complete system integration.

The intended operating sequence is: power-on and initialise peripherals; establish the local navigation state; follow the marked guidepath; check for an RFID checkpoint; monitor the forward obstacle distance; stop and alert if the obstacle condition is met; and update the remote interface when the network is available. This sequence describes the designed workflow. The validated result remains preliminary line-following operation and a Blynk interface test, while complete checkpoint

delivery, cloud logging and camera streaming require further integrated testing.

4.2 System Architecture / Block Diagram / Workflow

The architecture is organised into power, sensing, controller, actuation/safety, camera and IoT/data-logging subsystems, as illustrated in Fig. 4.1. The battery subsystem supplies the motor driver, while the LM2596 buck converter supplies regulated low-voltage power to the controller and electronics. Sensor outputs are processed by the ESP32, which controls the motor driver and transmits selected status information over Wi-Fi.

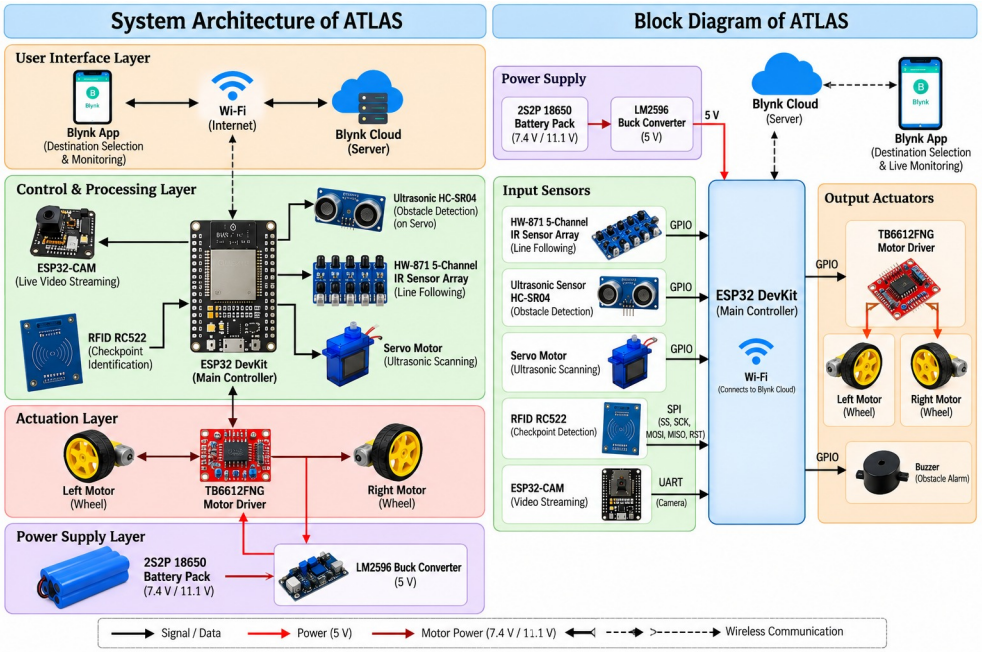


Figure 4.1: Project architecture and functional block diagram of ATLAS, showing the user-interface, control, sensing, actuation and power subsystems.

The implementation-level diagram combines the GPIO-level interconnection view with the working flowchart. It identifies the three IR sensor inputs, HC-SR04 trigger/echo connections, servo and buzzer outputs, TB6612FNG motor-driver signals, and the main navigation / obstacle-response sequence. The flowchart depicts the intended control logic, including line detection, end-of-line checking and obstacle response. It documents the design and demonstration workflow; it is not, by itself, quantitative evidence of successful completion of every operating case.

Diagram verification note: the supplied high-level architecture graphic labels a five-channel HW-871 IR array, whereas the implementation-level diagram and other project records show three line sensors. The diagrams also display more than one battery-voltage notation. Confirm the final sensor module and battery configuration against the actual prototype and firmware before final submission; the supplied graphics are reproduced as project-provided design artefacts.

4.3 Design Methodology

The design follows a modular sensing-decision-actuation approach. The ESP32 repeatedly reads the IR sensors, estimates the position of the guidepath relative to the sensor array, calculates a steering command and drives the left and right motors. A conceptual weighted line-position error for an

array of N sensors is:

$$e = \frac{\sum_{i=1}^{N} w_i s_i}{\sum_{i=1}^{N} s_i}, \tag{4.1}$$

where s_i represents the response of sensor i, and w_i is the corresponding position weight. The sign and magnitude of the error indicate whether the vehicle should steer left, travel approximately straight or steer right. The expression is a conceptual model; the supplied documents do not provide calibrated numerical weights or measured error values.

For a differential-drive platform, the left and right motor commands can be represented conceptually as:

$$v_L = v_0 - Ke, \qquad v_R = v_0 + Ke, \tag{4.2}$$

where v_0 is the nominal forward command and K is a steering gain. The exact controller gains and motor-speed values should be included only after verification against the final source code and test log.

The HC-SR04 measures the round-trip time of an ultrasonic pulse. The distance is estimated using:

$$d = \frac{c\,t}{2}, \tag{4.3}$$

where c is the speed of sound in air and t is the measured round-trip time. The division by two accounts for the outgoing and returning paths. A configured threshold is used to decide when the normal motion command should be overridden by a stop command; the final threshold value was not included in the supplied report material.

For RFID destination identification, the MFRC522 reader obtains the tag identifier and the controller compares it with the predefined tag-to-zone mapping. For IoT monitoring, selected vehicle states and events are formatted for transmission over Wi-Fi to Blynk and the Google Apps Script endpoint. The ESP32-CAM is treated as a visual monitoring extension rather than the primary navigation sensor.

4.4 Hardware / Software / Communication / Tools Selection

Table 4.1: Main hardware and software components

Component	Reported Specification	Function in the AGV
ESP32 development board	Main controller with Wi-Fi	Reads sensors, executes navigation and safety logic, and communicates with the monitoring layer.
IR line sensors	Three sensors: left, centre and right	Detect the black guidepath relative to the surrounding surface.
MFRC522 / RC522 RFID reader	HF RFID reader module	Reads checkpoint tags for warehouse-zone identification.

HC-SR04 ultrasonic sensor	Ultrasonic distance sensor	Measures forward distance to nearby obstacles.
SG90 servo motor	Servo-mounted scanning arrangement	Reorients the ultrasonic sensor for a wider scanning concept.
TB6612FNG motor driver	Dual-channel H-bridge driver	Drives two geared DC motors in a differential-drive arrangement.
TT geared DC motors	Two motors	Provide left and right wheel motion.
Battery subsystem	Four 18650 cells in 2S2P arrangement; nominal 7.4 V; 2S BMS	Supplies the mobile platform.
LM2596 buck converter	Regulated 5 V output as reported	Provides regulated low-voltage power to control electronics.
Buzzer	Audible status / warning	Provides a local warning indication.
ESP32-CAM	Camera module	Provides visual monitoring and intended delivery-image capture.
Blynk IoT	Dashboard platform	Displays selected AGV status information over Wi-Fi.
Google Apps Script / Sheets	Web endpoint and spreadsheet log	Intended to store timestamped operating events.
Arduino IDE / Embedded C-C++	Development environment	Used for ESP32 firmware development and module integration.

The ESP32 was selected because it combines embedded control with Wi-Fi connectivity in one development platform. The TB6612FNG provides dual motor channels for the two-motor drive arrangement. The MFRC522 supports checkpoint identification, while the HC-SR04 provides a low-complexity distance measurement method. Blynk provides a dashboard interface, and Google Apps Script / Sheets offers a simple event-recording path. Component selection and interfacing are based on the relevant manufacturer and platform documentation [17], [18], [19], [20], [21].

4.5 Design Constraints and Trade-offs

The main design trade-off is the use of a predefined physical guidepath instead of free navigation. This keeps sensing and control relatively simple and reduces the need for expensive global-localisation equipment, but the vehicle cannot independently select an alternate route around a blocked aisle. Three IR sensors are sufficient for a basic demonstration but provide less lateral information than

a larger sensor array. RFID checkpoint identification gives semantic destination information but depends on tag position, orientation and reader range. Ultrasonic sensing is inexpensive but its response can be affected by target geometry and environmental conditions.

The battery, BMS and buck converter provide a compact portable power arrangement, but runtime and current margins require measurement under the actual motor load. Wi-Fi enables remote monitoring but should not be part of the critical local navigation loop. The camera improves visibility but introduces additional power, network and privacy considerations. The present prototype therefore prioritises low cost, modularity and demonstration feasibility over industrial-level throughput, redundancy or safety certification.

5. IMPLEMENTATION AND DEVELOPMENT

5.1 Implementation Details

The prototype was assembled on a layered mobile chassis using two geared DC motors in a differential-drive arrangement. The ESP32 acts as the central controller, while the line sensors are placed to observe the prepared black guidepath. The RFID reader is provided for checkpoint detection using the MFRC522 interface, and the ultrasonic sensor is positioned at the front of the vehicle. The SG90 servo supports the ultrasonic scanning concept. The ESP32-CAM is mounted as an additional visual-monitoring subsystem. The ESP32 and MFRC522 interfaces are based on their respective device documentation [18], [19].

The power subsystem uses four 18650 cells in a 2S2P arrangement, a 2S battery-management system and an LM2596 buck converter. As described in the technical paper, the battery supplies the motor driver and the buck converter provides regulated power for the controller and low-voltage electronics. The TB6612FNG is used as the dual-channel motor driver in the differential-drive stage [17]. Software is organised around modular drivers for line sensing, RFID, the motor driver, ultrasonic sensing, Wi-Fi communication and camera monitoring. The documented monitoring and event-logging platforms are Blynk and Google Apps Script / Sheets [20], [21].

5.2 Module-wise / Phase-wise Development

5.2.1 Mechanical Platform and Drive

A layered chassis carries the control electronics, sensors, power subsystem and camera. Two TT geared DC motors provide differential drive. The motor driver varies the left and right motor commands to correct the vehicle's position relative to the guidepath.

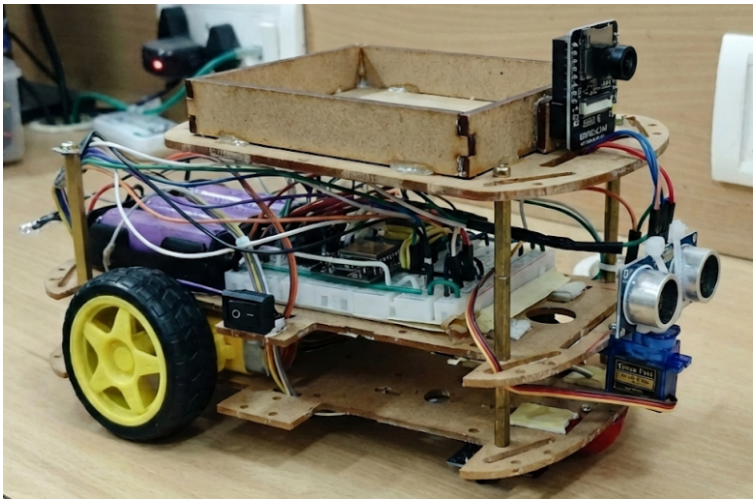


Figure 5.1: Developed AGV prototype used for preliminary functional validation.

5.2.2 Line-following Subsystem

Three IR sensors observe the black guidepath and provide the input required for steering. Sensor calibration and black/white detection were identified as issues requiring attention in the project

logbook. The supplied technical paper reports that the line-following subsystem was functionally demonstrated on the developed guidepath; it does not provide a numerical path accuracy or speed measurement.

5.2.3 RFID Checkpoint Subsystem

The MFRC522/RC522 reader is integrated for reading tags placed near predefined warehouse zones. The controller is intended to compare the tag identifier against stored destination information. The project logbook reports RFID hardware integration as complete, while destination logic remains under system testing. Read-range and reliability measurements were not supplied.

5.2.4 Obstacle Detection and Servo Subsystem

The HC-SR04 sensor is used to estimate the distance to an object in front of the AGV. A servo-mounted arrangement supports scanning beyond a single forward direction. The logbook reports that obstacle-sensor integration is complete, while threshold setting and field-response testing remain in progress. The present report therefore does not claim validated obstacle avoidance around an object.

5.2.5 IoT Monitoring and Camera Subsystems

The ESP32 Wi-Fi interface is used to support the Blynk monitoring dashboard and a Google Apps Script / Google Sheets logging architecture. A Blynk interface screenshot is included in Fig. 5.2; the technical paper clarifies that this demonstrates dashboard connectivity and status reporting, not successful end-to-end delivery validation. The ESP32-CAM is included as a visual-monitoring extension, but integrated streaming validation remains ongoing.

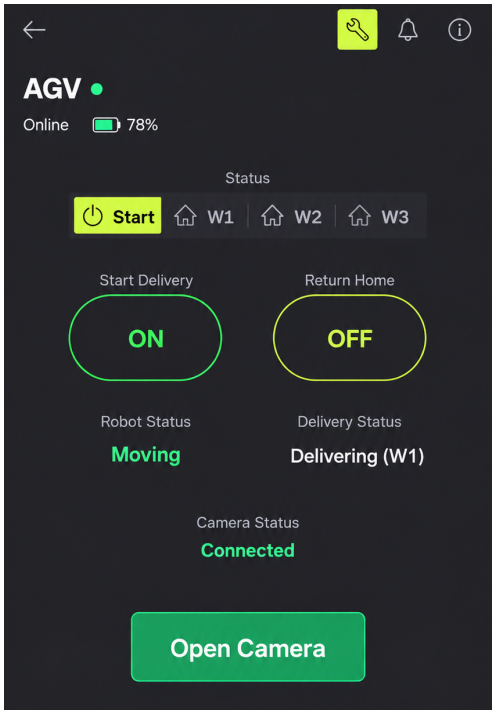


Figure 5.2: Blynk IoT dashboard interface documented for AGV status monitoring. The screenshot demonstrates the interface test only.

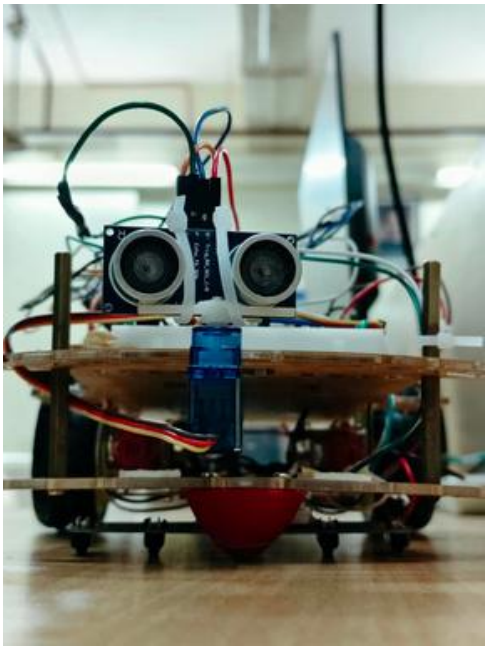


Figure 5.3: Front view of the assembled AGV.

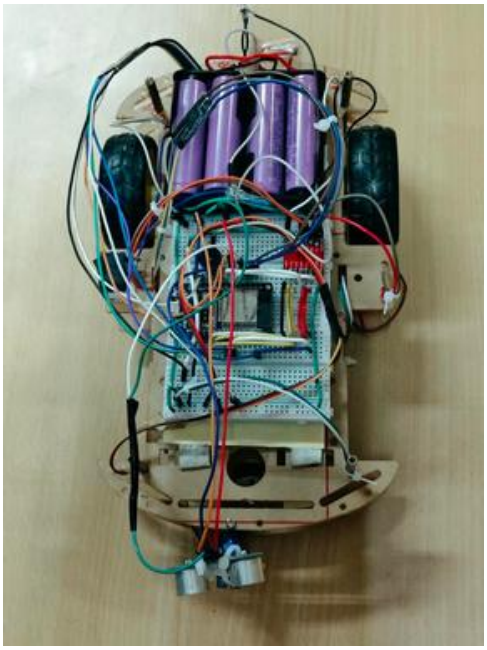


Figure 5.4: Top view of the AGV hardware arrangement.

5.2.6 Exhibition Setup and Prototype Demonstration

The following photographs were captured during the project exhibition working phase. They document the AGV carrying a demonstration load, the vehicle positioned on the black guide track, the front-mounted sensing arrangement, and the project demonstration environment with the report / poster display. These photographs provide visual evidence of the assembled prototype and its exhibition setup. They are not used to infer numerical speed, payload capacity, navigation accuracy or obstacle-avoidance performance.

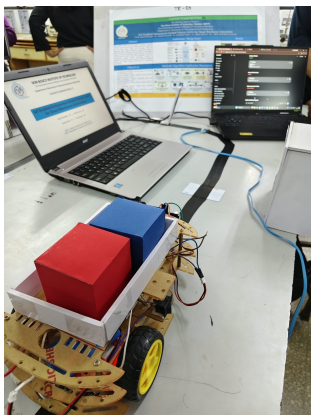


Figure 5.5: ATLAS carrying the demonstration payload during the exhibition setup.



Figure 5.6: Side view of the AGV placed on the black guide track at the exhibition.

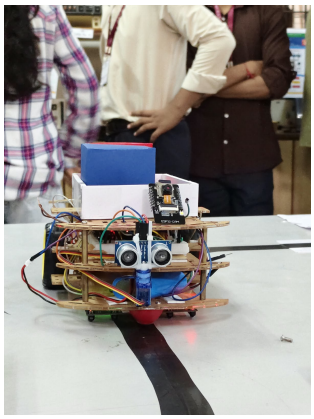


Figure 5.7: Front view showing the ultrasonic sensing and upper camera arrangement.

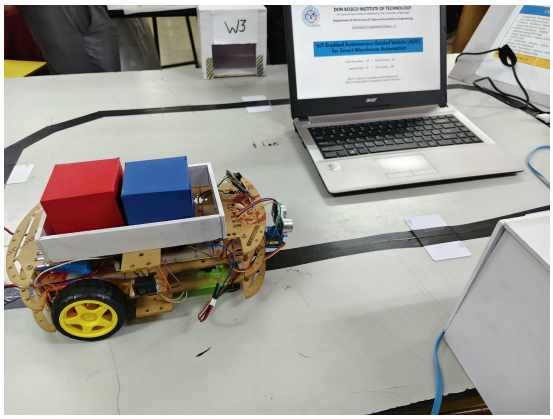


Figure 5.8: AGV and demonstration track with the project presentation material.

5.3 Integration and Troubleshooting

The project logbook identifies sensor calibration and multi-module integration as key challenges. It also states that detailed completed corrective actions should be updated from the team's actual test log. The table below therefore records only the documented observations and the validation actions required; it does not invent undocumented fixes.

Table 5.1: Documented integration challenges and follow-up actions

Challenge / Observation	Corrective Action / Follow-up	Outcome
IR sensor calibration and black/white detection required attention.	Calibrate sensor position and detection threshold; record the actual final settings and repeatability results in the test log.	Line-following functionality was demonstrated; no numerical accuracy data were supplied.

RFID destination logic remained under testing after hardware integration.	Complete tag-to-zone mapping and record successful reads, failed reads, orientation and read distance.	RFID hardware integration reported complete; destination logic validation remains pending.
Obstacle sensor integration was reported complete, but threshold / field testing remained in progress.	Set and document the obstacle threshold; test repeated approaches at known distances and verify motor-stop behaviour.	Integrated obstacle-response validation not supplied.
Full multi-module validation remained incomplete.	Perform staged module tests followed by an end-to-end route test with event logging and camera monitoring.	Full end-to-end delivery validation and quantitative performance measurements were not supplied.

5.4 Cost / Resource Summary

The available component-cost record contains estimated prices rather than verified receipt amounts. The listed line items add up to approximately INR 2,680, while the earlier overall estimate was INR 2,000–2,500. The battery entry also needs confirmation because the cost record refers to a 2-cell pack, whereas the technical documentation describes four 18650 cells in a 2S2P arrangement. The amounts below are therefore retained as provisional estimates; the team must reconcile quantities, reused components, discounts and actual receipts before treating the total as the final project expenditure.

Table 5.2: Estimated bill of materials based on the project poster

Item / Resource	Quantity	Approx. Cost (INR)	Remarks
ESP32 development board	1	450	Main controller.
ESP32-CAM module	1	700	Visual-monitoring extension.
IR line sensors	3	90	Left, centre and right sensing.
HC-SR04 ultrasonic sensor	1	75	Obstacle-distance measurement.
MFRC522 / RC522 RFID reader	1	80	Checkpoint identification.
RFID cards (13.56 MHz)	3	45	Demonstration tags.
SG90 servo motor	1	100	Ultrasonic scanning concept.
TB6612FNG dual motor driver	1	150	Drives the two geared motors.
BO / TT geared motors	2	100	Differential-drive propulsion.
Wheels	2	60	Drive wheels.

Item / Resource	Quantity	Approx. Cost (INR)	Remarks
Caster wheel	1	10	Passive support wheel.
Buzzer	1	20	Audible status / warning.
18650 battery cells / 2S2P pack	4 cells	190	Poster price is shown for a 2-cell pack; technical paper describes 4 cells. Reconfirm pack cost.
LM2596 buck converter	1	50	Regulated low-voltage supply.
2S battery management system (BMS)	1	80	Battery protection.
Chassis	1	180	Mechanical platform.
Servo mounting bracket	1	100	Servo mounting.
Mounting screws	1 set	50	Mechanical assembly.
Connecting wires and connectors	1 set	150	Electrical interconnections.
Sum of listed poster line items		**2,680**	

The sum above is the arithmetic total of the listed estimated line items, not a verified amount paid. The final total should be updated from actual receipts and reconciled with the confirmed battery configuration.

6. TESTING, RESULTS AND VALIDATION

6.1 Test Plan / Validation Method

The validation plan is organised around the functional requirements defined in Section 2.4. Only the results explicitly supported by the supplied report and logbook are identified as demonstrated. Where a test result was not supplied, the observed-result field states that fact.

Table 6.1: Test plan and current validation status

Test / Criterion	Method	Expected Result / Target	Observed Result	Status
Line-following navigation	Run the AGV over the prepared black guidepath and observe tracking through turns / straight segments.	Follow the predefined route using three IR sensors.	Functional line following on the developed guidepath is reported. No numerical accuracy or speed data supplied.	Demonstrated (qualitative)
IR sensor response	Place the sensor array over black and white surfaces and record output states.	Consistent distinction between guidepath and background.	Calibration and black/white detection required attention; detailed raw readings not supplied.	Partially documented.
RFID checkpoint identification	Present a known tag at a checkpoint and verify the detected UID / destination.	Correctly associate a valid tag with its stored warehouse zone.	RFID hardware integration reported complete; destination logic remained under system testing.	In progress.
Ultrasonic obstacle response	Place an obstacle at known distances and check the stop response at the configured threshold.	Stop the motors when the obstacle condition is met.	Sensor integration reported complete; threshold and field testing remained in progress.	In progress.

Test / Criterion	Method	Expected Result / Target	Observed Result	Status
Blynk dashboard	Connect the controller / dashboard and check displayed status fields.	Display selected online, battery, status and destination fields.	Dashboard interface screenshot and connectivity/status reporting are documented. Not proof of end-to-end delivery.	Interface test documented
Google Sheets event logging	Trigger a defined event and verify timestamp, tag ID and status in the sheet.	Record selected events with valid timestamps.	Architecture is described; no end-to-end log output was supplied.	Not validated in supplied evidence
ESP32-CAM visual monitoring	Connect the camera and verify live stream / delivery snapshot.	Provide remote visual monitoring and image capture.	Module is included in the design; integrated streaming validation is ongoing.	In progress.
Complete route / delivery	Execute the route from start to destination with sensor, RFID, camera and IoT functions integrated.	Complete a repeatable delivery with traceable event record.	No full end-to-end delivery validation was supplied.	Pending.

6.2 Results and Performance Analysis

The principal functional result supported by the supplied technical paper is preliminary line-following operation on the developed guidepath. This indicates that the three-sensor IR arrangement and differential-drive platform can acquire and track the prepared path under the tested prototype conditions. The available evidence is qualitative; no route-completion percentage, tracking error, speed, repeatability or payload data were supplied.

The Blynk dashboard screenshot in Fig. 5.2 shows a monitoring interface with online status, battery indication, start / destination controls, vehicle status and camera status fields. The paper explicitly states that the screenshot represents dashboard connectivity and status reporting only. It does not prove successful delivery, accurate battery measurement under load or complete sensor-to-cloud integration.

The prototype photograph in Fig. 5.1 confirms the physical assembly of a layered mobile platform

with motors, sensors and the camera subsystem. The exhibition photographs in Figs. 5.5–5.8 further document the vehicle with a demonstration payload, its placement on the marked track, the front sensor arrangement and the exhibition presentation setup. They strengthen the visual record of prototype assembly and demonstration. They do not provide measured payload capacity, route-completion rate or obstacle-avoidance reliability. The logbook reports RFID hardware integration and ultrasonic sensor integration, but also records destination-logic testing and obstacle threshold / field validation as ongoing. Consequently, the system should be reported as a functional proof of concept with preliminary line-following validation and exhibition-stage visual evidence, rather than a quantitatively validated industrial warehouse AGV.

For quantitative evaluation, future tests should record the number of successful line-following runs, route length, travel time, line-loss events, RFID reads / attempts, obstacle detection distance and stop latency, Blynk update latency, Google Sheets record completeness and battery operating time. Each result should be repeated under controlled conditions and reported with the test setup.

6.3 Stakeholder Feedback and Validation

The supplied logbook records desk research on 15 October 2026 for three potential logistics stakeholders, and explicitly states that no direct community or industry interaction has been confirmed. No interview, survey, site visit, permission letter or stakeholder evaluation was supplied. Therefore, direct stakeholder feedback cannot be reported.

Table 6.2: Stakeholder feedback and validation status

Stakeholder	Feedback Method	Key Feedback	Action / Interpretation
Potential logistics stakeholders identified in the logbook: ColdStar Logistics Pvt. Ltd., TCF Logistics Pvt. Ltd. and LP Logiscience LLP.	Desk research of publicly available websites on 15 Oct. 2026; no direct interaction.	No interview feedback or operational requirements were collected. Public information was used only to identify potential relevance.	Arrange and document a real consultation, visit or written response before claiming community validation.

6.4 Achievement of Project Objectives

Table 6.3: Objective-wise achievement status

Objective	Status	Evidence / Result	Remarks
Design and assemble a compact AGV prototype.	Achieved (prototype assembly)	Developed layered chassis with two-motor differential drive and integrated electronics.	Industrial payload and endurance validation not supplied.

Implement three-sensor line following.	Achieved (qualitative)	Preliminary line-following functionality demonstrated on the developed guidepath.	Accuracy and speed values not reported.
Identify warehouse checkpoints using RFID.	Partially achieved	RFID hardware integration reported complete.	Tag-to-zone logic remained under system testing; read reliability not reported.
Detect obstacles and provide a stop response.	Partially achieved	HC-SR04 integration reported complete.	Threshold setting and field-response tests remained in progress.
Provide Blynk monitoring and Google Sheets logging.	Partially achieved	Blynk dashboard interface / status reporting screenshot documented.	Google Sheets end-to-end event logging not validated in the supplied evidence.
Provide ESP32-CAM visual monitoring.	Partially achieved	ESP32-CAM included in system architecture and mounted on prototype.	Live streaming and image-capture validation ongoing.

7. SUSTAINABILITY, SOCIETAL IMPACT AND ENGINEERING CONSIDERATIONS

7.1 Sustainability Assessment

The project supports the educational and innovation-oriented intent of SDG 9 (Industry, Innovation and Infrastructure) by developing a low-cost embedded automation prototype and applying electronics, sensing, control and wireless communication to a practical material-handling problem. SDG 9 is relevant because it encourages resilient infrastructure, sustainable industrial development and innovation. The project is a small-scale proof of concept and is not evidence of industrial deployment or increased industrial productivity.

The project also relates to SDG 12 (Responsible Consumption and Production), which encourages efficient resource use, waste prevention, responsible management of materials and environmentally sound handling of waste. A reusable prototype with replaceable modules can support learning and repair rather than unnecessary disposal. In a warehouse context, better tracking of material movement could potentially reduce avoidable handling and process inefficiency, but this benefit has not been measured in the present work.

Sustainability benefits in a real warehouse would depend on energy consumption, battery life, maintenance, component replacement, operational throughput and the amount of manual handling actually displaced. None of these indicators has been measured for a warehouse deployment. Battery cells and electronic modules should be handled, charged, maintained and disposed of according to applicable safety and e-waste procedures. SDG 12 alignment is therefore a potential direction, not a verified reduction in resource use.

7.2 Societal Relevance, Inclusiveness and Accessibility

The project addresses the general need for more organised movement of goods in warehouse environments. A guided vehicle may support workers by handling repetitive movement along a fixed route while allowing an operator to monitor selected vehicle states. A low-cost demonstrator can also be useful for technical education and for introducing students to embedded automation and IoT-based monitoring.

The current system assumes a prepared route and a controlled environment. It does not yet establish usability for workers with different accessibility needs, nor has it been evaluated with warehouse personnel. Any future deployment should consider clear visual and audible indications, accessible emergency controls, safe operating speed, route markings, safe separation from pedestrians and training for operators. No direct community feedback has been collected to date.

7.3 Ethical, Safety and Professional Responsibility

Engineering responsibility requires accurate reporting of system performance and clear separation between demonstrated functions and planned features. The report does not claim obstacle avoidance, full RFID delivery or end-to-end IoT validation when the supplied records describe those functions

as ongoing. Stakeholder desk research is distinguished from interviews, site observations or endorsements.

The camera subsystem requires permission and appropriate data handling before use in a real warehouse. Image capture should be limited to the purpose of vehicle monitoring, access should be restricted and unnecessary personal data should not be collected. Battery protection, wiring insulation, mechanical guarding and supervised testing are essential. Since the project is not an industrially certified safety system, it should not be operated in a live warehouse with workers or forklifts without a separate risk assessment and appropriate engineering safeguards.

7.4 Complex Engineering / Problem-Solving Evidence

The project combines power electronics, embedded control, sensor interfacing, motor actuation, wireless communication, RFID identification and camera monitoring in one mobile platform. These functions have different electrical, timing and software requirements. The design therefore requires decisions about power distribution, sensor placement, differential motor control, local versus cloud processing, communication reliability and physical integration. Sensor calibration, RFID destination logic and staged integration were identified as areas requiring continued work in the logbook.

The complexity is addressed through a layered architecture and modular firmware approach. Functional line-following tests provide preliminary evidence for the navigation subsystem, while other functions are being integrated and validated. This evidence-based description avoids treating a proposed block or workflow as a completed performance result.

7.5 Community Impact / Potential Impact

The potential community impact is a demonstration of how a low-cost guided vehicle could support repetitive material transport and provide basic visibility of vehicle status. Relevant impact indicators for future evaluation include operator time per delivery, manual intervention count, successful checkpoint identification, obstacle-stop response, route-completion rate and accuracy of event logs. These indicators require controlled trials and direct stakeholder assessment. At present, the project demonstrates preliminary navigation and a monitoring interface, not measured productivity gains or accident reduction.

8. CONCLUSION, REFLECTION AND FUTURE WORK

8.1 Conclusion

This project presents the design and preliminary development of an IoT-enabled AGV proof of concept for structured warehouse material movement. The prototype combines ESP32-based control, three-IR-sensor line following, RFID checkpoint hardware, ultrasonic obstacle sensing, a two-motor differential-drive platform, Wi-Fi monitoring architecture and an ESP32-CAM visual-monitoring extension. The supplied technical report confirms preliminary functional demonstration of line following and documents a Blynk dashboard interface test.

The work establishes a modular basis for integrating navigation, destination identification, obstacle awareness and remote monitoring on an economical embedded platform. However, the supplied evidence does not establish complete end-to-end delivery validation, quantitative performance, direct stakeholder acceptance or readiness for industrial operation. The project therefore addresses the selected need at proof-of-concept level and identifies specific validation work required before practical deployment can be considered.

8.2 Limitations

The AGV relies on a predefined physical guidepath and cannot dynamically re-route around a blocked aisle. Quantitative values for line-following accuracy, route speed, payload, RFID reading range, obstacle-stop latency, battery endurance and wireless latency were not supplied. RFID destination logic and obstacle threshold / field testing were still in progress in the logbook. Google Sheets event logging and ESP32-CAM streaming did not have documented end-to-end validation. No direct stakeholder interview, site visit or formal feedback was recorded. The prototype is not a certified industrial safety device.

8.3 Future Improvements and Scalability

The following improvements are recommended:

1. Complete RFID tag-to-zone logic and evaluate read reliability over different tag positions and orientations.
2. Tune the ultrasonic threshold and validate stop / resume behaviour using repeatable distance trials; develop a tested obstacle-bypass strategy only after safety review.
3. Complete Google Sheets event logging and ESP32-CAM streaming, then measure communication latency and record completeness.
4. Collect quantitative line-following data and tune a proportional or PID controller using measured lateral error.
5. Add a payload measurement mechanism, improved chassis guarding and a safe charging / docking arrangement.
6. Explore multi-AGV coordination, route optimisation, LiDAR / UWB / visual localisation and

warehouse-management-system integration as future research directions.

7. Conduct a documented consultation or controlled demonstration with at least one external warehouse stakeholder and record its feedback.

8.4 Team Reflection and Learning Outcomes

The project provided experience in translating a warehouse material-handling use case into an embedded-system architecture. The team gained practical exposure to ESP32 programming, IR sensor calibration, line-following logic, RFID interfacing, ultrasonic distance measurement, differential-drive motor control, power regulation and IoT monitoring concepts. The literature review and project presentations also developed technical communication and documentation skills.

The work highlighted the importance of staged testing, recording corrective actions and separating a successful subsystem demonstration from full system validation. It also reinforced the need to consider battery safety, camera privacy, stakeholder consent and the social context of a community-oriented engineering project. The team should update this reflection with specific individual learning examples before submission.

8.5 Individual Contribution of Team Members

The following table presents the individual contributions of all four team members towards the design, development, implementation, testing and documentation of the ATLAS autonomous guided vehicle. The contribution percentages represent the approximate distribution of responsibilities undertaken by each member.

Table 8.1: Individual contribution of team members

Student Name	**Major Contribution(s)**	**Tools / Tasks Handled**	**Approx. Contribution (%)**
Yash Khanvilkar	Developed the obstacle-detection and path-changing algorithms. Worked on obstacle-avoidance logic and preparation of the project report.	Obstacle-detection programming; path-changing algorithms; obstacle-avoidance logic; code testing; project report preparation.	25%
Rohit Kumar	Developed major software components, including the Blynk interface and Google Docs integration. Handled code compilation, ESP32-CAM integration and final software development.	Blynk IoT interface; Google Docs integration; ESP32-CAM integration; code compilation; software integration; final coding.	25%

Harshit Patil	Handled major hardware development and electrical connections. Integrated the electronic components, debugged hardware issues and implemented the robot's line-following functionality.	ESP32 DevKit; TB6612FNG motor driver; HW-871 IR sensor array; motor and power connections; RFID; ultrasonic sensor; servo motor; hardware integration; line-following implementation.	25%
Tirth Shetty	Assisted with hardware development, component verification and rechecking of system connections. Prepared the progress-seminar presentation and final project poster.	Hardware assistance; component rechecking; system verification; progress-seminar PPT; final exhibition poster; presentation materials.	25%
		Total Contribution	**100%**

REFERENCES

[1] I. Kubasakova, J. Kubanova, D. Benco, and D. Kadlecova, "Implementation of Automated Guided Vehicles for the Automation of Selected Processes and Elimination of Collisions between Handling Equipment and Humans in the Warehouse," *Sensors*, vol. 24, p. 1029, 2024.

[2] K. Ellithy, M. Salah, I. S. Fahim, and R. Shalaby, "AGV and Industry 4.0 in warehouses: a comprehensive analysis of existing literature and an innovative framework for flexible automation," *The International Journal of Advanced Manufacturing Technology*, 2024.

[3] K. Staffenova, M. Rakyta, and V. Binasova, "The use of automated guided vehicles in the internal logistics of the production company," *Transportation Research Procedia*, vol. 74, pp. 458–464, 2023.

[4] J. Alam, D. S. Rajput, and S. Baraskar, "Review and Challenges of Automated Guided Vehicle," *International Journal of Research and Analytical Reviews*, vol. 10, no. 2, 2023.

[5] G. T. S. Ho, Y. M. Tang, E. K. H. Leung, and P. H. Tong, "Integrated reinforcement learning of automated guided vehicles: dynamic path planning for smart logistics and operations," *Transportation Research Part E*, 2025.

[6] K. Wang, W. Liang, H. Shi, J. Zhang, Q. Wang et al., "Optimal time reuse strategy-based dynamic multi-AGV path planning method," *Complex & Intelligent Systems*, vol. 10, pp. 7089–7108, 2024.

[7] G. S. Nage, "Design and Methodology of Automated Guided Vehicle," *International Journal of Scientific Development and Research*, vol. 7, no. 10, 2022.

[8] R. Bourafa and P. Siegfried, "A Review of the Automated Guided Vehicle Systems: Dispatching Systems and Navigation Concept," *Automobile Transport*, vol. 52, 2023.

[9] M. Aizat, A. Azmin, and W. Rahiman, "A Survey on Navigation Approaches for Automated Guided Vehicle Robots in Dynamic Surrounding," *IEEE Access*, 2023.

[10] A. Chakraborty, "The Future of Logistics: SAP EWM and Autonomous Vehicles in Smart Warehouse Automation," *Transactions on Engineering and Computing Sciences*, vol. 13, no. 01, 2025.

[11] B. Hadisujoto, A. N. Rabbani, N. K. H. Fernandez, K. Saptaji, and D. Wibowo, "Development of 3D Printed Autonomous Warehouse Robot Using Mecanum Wheel and Robot Arm," *Journal of Applied Science and Advanced Technology*, 2023.

[12] Q. Pan, "Design and Analysis of an Autonomous Warehouse Robot System with 6-DOF Manipulator," in *Proc. 2023 International Conference on Machine Learning and Automation*, 2023.

[13] Z. Zhao, A. Mazen, Y. Bazzi, B. Wang, and S. Li, "Intelligent Warehouse Robot for Classifying, Sorting, and Transporting Goods' Boxes," in *Proc. IEEE IECON*, 2024.

[14] D. Pandit, C. Kalbhor, S. Chitte, and D. Chandel, "RFID-Based Autonomous Line Following Robot for Smart Warehouse Inventory Management," *International Journal of Innovative Research in Technology (IJIRT)*, 2026.

[15] A. Sahu, M. D. Amaan, M. S. S. Kiran, and N. K. Dewangan, "Autonomous Warehouse Robot with Priority-Based Sorting," *Journal of Emerging Trends and Novel Research (JETNR)*, 2026.

[16] S. Katwate, K. Yadav, T. Deshpande, S. Shaikh, and J. A. Shaikh, "Design & Development of Automated Guided Vehicle for Small Scale Industry," *International Journal of Scientific Research in Engineering and Management*, vol. 08, no. 04, 2024, doi: 10.55041/IJSREM31407.

[17] Toshiba Electronic Devices & Storage Corporation, "TB6612FNG Brush Motor Driver IC Datasheet," 2014.

[18] Espressif Systems, "ESP32 Series Datasheet," version 5.3, 2026.

[19] NXP Semiconductors, "MFRC522 Contactless Reader IC Datasheet."

[20] Blynk, "Blynk IoT Documentation," Blynk Inc., accessed Oct. 2026.

[21] Google, "Google Apps Script Documentation." [Online]. Available: `https://developers.google.com/apps-script`

APPENDICES

Appendix A – Stakeholder Interaction / Community Evidence

The project logbook contains a desk-research record dated 15 October 2026 for three potential stakeholders. It explicitly states that no direct interaction, interview, site visit or stakeholder feedback has been confirmed. The public information below is therefore contextual evidence only and must not be described as community consultation.

Table 8.2: Publicly available stakeholder information recorded in the logbook

Organization	Desk-research note	Public source recorded in logbook
ColdStar Logistics Pvt. Ltd.	Public company information describes temperature-controlled warehousing, inventory management, shipment visibility and monitoring. No staff member was contacted.	`https://coldstarlogistics.com/`
TCF Logistics Pvt. Ltd.	Public website lists warehouse space and locations including Bhiwandi and JNPT. No operational feedback was collected.	`https://tcflogistics.in/`
LP Logiscience LLP	Public contact information lists warehouse facilities in Bhiwandi, Thane district. No site visit or interview was conducted.	`https://lplogiscience.com/support/contact-us/`

Evidence still required for final CEP submission: attach at least one genuine dated stakeholder interaction artefact if required by the course template, such as an organisation email, acknowledgement, signed visit record, meeting note or feedback form. No such document was included in the uploaded files. Do not fabricate a letter, signature or feedback statement.

Appendix B – Testing / Validation Evidence

The following evidence is available in the supplied technical paper and project materials:

- Preliminary functional demonstration of line following on the developed guidepath.
- Photograph of the assembled AGV prototype (Fig. 5.1).
- Exhibition-phase photographs showing the demonstration load, track placement, front sensing arrangement and project presentation setup (Figs. 5.5–5.8).
- Project-provided system architecture, functional block diagram, implementation-level GPIO diagram and working flowchart (Figs. 4.1 and **??**).
- Blynk dashboard screenshot showing the monitoring interface (Fig. 5.2); this proves interface

connectivity/status reporting only.

Detailed quantitative test sheets, raw IR readings, RFID read logs, ultrasonic threshold trials, Google Sheets event records, camera-stream screenshots and battery endurance measurements were not included in the supplied files. Add these artefacts after tests are completed.

Appendix C – Technical Documentation

Power architecture

The documented battery subsystem uses four 18650 lithium-ion cells connected in a 2S2P arrangement, giving a nominal pack voltage of approximately 7.4 V. A 2S battery-management system (BMS) is included for cell protection and pack-level monitoring / protection functions. The battery subsystem supplies the motor-drive stage, while an LM2596 DC–DC buck converter steps the battery voltage down to a regulated 5 V rail for the controller and compatible low-voltage electronics. The motor supply and regulated logic supply should share a suitable common ground, and the wiring, connector and converter current ratings must be verified against the actual load. The pack must be charged using a compatible 2S charging arrangement; individual cells must not be charged directly in an improvised configuration. The final cell count, wiring polarity and measured output voltages should be checked against the assembled prototype before submission.

Control architecture

ESP32 main controller; three IR sensors for line tracking; MFRC522/RC522 RFID reader for checkpoint identification; HC-SR04 ultrasonic sensor with SG90 servo scanning concept; TB6612FNG motor driver; two TT geared DC motors; buzzer; ESP32-CAM; Blynk IoT; Google Apps Script and Google Sheets.

Firmware structure (conceptual):

1. Initialise ESP32, sensors, RFID, motor driver, Wi-Fi and camera subsystem.
2. Read IR sensor states and calculate the line-position error.
3. Determine left / straight / right steering and update motor commands.
4. Read ultrasonic distance and apply the configured local stop condition.
5. Check RFID tags and map valid identifiers to predefined warehouse zones.
6. Send selected status / event information to Blynk and Google Sheets when communication is available.
7. Provide ESP32-CAM visual monitoring and mark route completion when the assigned sequence is complete.

This is a functional description of the architecture, not a substitute for the final verified firmware listing. Insert the final source-code repository link or selected code extracts after the team has frozen and tested the code.

Appendix D – Project Demonstration / Repository Links

Item	Link / Status
Demonstration Video	Watch AGV demonstration on YouTube
GitHub	Our Complete Source code is available here
Dataset / Data Repository	NA
ESP32-WROOM-32 Datasheet	Open ESP32-WROOM-32 datasheet (PDF)
TB6612FNG Motor Driver Datasheet	Open TB6612FNG datasheet (PDF)
TCRT5000 IR Sensor Datasheet	Open TCRT5000 datasheet (PDF)
MFRC522 RFID Reader Datasheet	Open MFRC522 datasheet (PDF)
AI-Thinker ESP32-CAM Datasheet	Open ESP32-CAM datasheet (PDF)
LM2596 Buck Converter Datasheet	Open LM2596 datasheet (PDF)
Other Evidence	NA

Appendix E – Course Outcomes, CO-PO-PSO Mapping and Evidence of CO Achievement

Course Outcome evidence index

Table 8.3: Course outcomes and project evidence

Course Outcome	Project Evidence / Artefact	Report Section / Appendix Reference
CO1: Identify community needs, stakeholder requirements, Sustainable Development Goals (SDGs), and relevant Knowledge and Attitude Profiles (KAPs) associated with the selected project. (Remember – Factual Knowledge)	Problem context, stakeholder desk-research record and SDG / KAP analysis.	Sections 1.2–1.3, 3.3–3.4; Appendix A.
CO2: Explain the social, environmental, technological, economic, and ethical factors influencing the development and implementation of community-oriented engineering solutions. (Understand – Conceptual Knowledge)	Existing-solutions comparison, design trade-offs, sustainability and safety discussion.	Sections 2.2–2.3, 4.5 and Chapter 7.
CO3: Apply appropriate engineering tools, techniques, methodologies, and technologies to develop and improve solutions addressing identified community needs. (Apply – Procedural Knowledge)	ESP32-based architecture, IR navigation, RFID / ultrasonic hardware, motor driver and firmware structure.	Chapters 4–5; Appendix C.

CO4: Analyse system performance, stakeholder feedback, implementation challenges, and operational constraints to determine the effectiveness of the proposed solution. (Analyse – Conceptual and Procedural Knowledge)	Test plan, preliminary line-following observation, dashboard evidence and integration challenge table.	Sections 5.3 and 6.1–6.3; Appendix B.
CO5: Evaluate the functionality, sustainability, reliability, inclusiveness, and societal impact of the developed solution using appropriate technical and community-benefit criteria. (Evaluate – Metacognitive Knowledge)	Objective achievement table, limitations and potential-impact assessment.	Sections 6.2–6.4, Chapter 7 and Section 8.2.
CO6: Design and validate an innovative, sustainable, and socially relevant engineering solution that addresses community needs while demonstrating responsible professional attitudes and relevant KAP attainment. (Create – Metacognitive Knowledge)	System architecture, prototype photographs, validation evidence and KAP reflection.	Chapters 3–8; Appendices B and C.

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1		2				3	1		1		1		2
CO2	1	2				3	2						2
CO3	3	2	3	1	3			2		1	1	3	3
CO4	2	3		3	2	2		1				2	3
CO5		2	2	2		3	2		1	1		1	3
CO6	2	2	3	2	2	3	2	3	2	2	2	3	3
Total	8	13	8	8	7	14	7	6	4	4	4	9	16
Avg.	2.00	2.17	2.67	2.00	2.33	2.80	1.75	2.00	1.33	1.33	1.33	2.25	2.67

(1 = Low, 2 = Medium, 3 = High, Blank = No Mapping)

CO-PO-PSO Mapping Justification

Table 8.4: CO-PO-PSO mapping justification

CO	Mapped PO(s) / PSO(s)	Justification for Mapping
CO1	PO2, PO6, PO7, PO9, PO11; PSO2	Identification of community needs, stakeholder requirements, SDGs and KAPs supports problem analysis and understanding of societal / sustainability context, with supporting links to ethics, communication and self-learning.
CO2	PO1, PO2, PO6, PO7; PSO2	Analysis of social, environmental, technological, economic and ethical factors requires conceptual engineering understanding and consideration of the wider impact of engineering solutions.
CO3	PO1, PO2, PO3, PO4, PO5, PO8, PO10, PO11; PSO1, PSO2	Application of engineering knowledge, tools and technologies to develop the solution directly supports design / development, tool usage and project-based technical implementation.
CO4	PO1, PO2, PO4, PO5, PO6, PO8; PSO1, PSO2	Performance analysis, testing, stakeholder feedback and troubleshooting require systematic investigation, interpretation of evidence and practical problem solving.
CO5	PO2, PO3, PO4, PO6, PO7, PO9, PO10; PSO1, PSO2	Evaluation of functionality, reliability, sustainability, inclusiveness and societal impact integrates technical performance with ethical and community-benefit considerations.
CO6	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11; PSO1, PSO2	Design and validation of an innovative, sustainable and socially relevant solution integrates technical design, teamwork, communication, ethics, project management and lifelong-learning attributes.